「Def. Let $A \in M_{m \times n}(F)$ be given where $F = \mathbb{R}$ or $\mathbb{C}$. Define the supremum norm of A

$$\|A\|_p \overset{\text{def}}{=} \sup_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p} \overset{(*)}{=} \sup_{\|x\|_p = 1} \|Ax\|_p$$

where $\|\cdot\|_p$ is the P-norm, that is,

$$\|\cdot\|_p : F^r \to \mathbb{R} ; (a_i)_{i=1}^{n} \mapsto \left(\sum_{i=1}^{n} |a_i|^p\right)^{\frac{1}{p}}.$$

The equality $(*)$ is given by the fact that; for any non-zero vector x, then

$$\frac{\|Ax\|_p}{\|x\|_p} = \left\|A \cdot \frac{x}{\|x\|_p}\right\|_p = \|Ax'\|_p, \text{ where } \|x'\| = 1.$$

In fact, the value $\|A\|_p$ is the smallest value C such that

$$\|Ax\|_p \leq C \cdot \|x\|_p \text{ for all non-zero } x \in F^n$$

That is, given $A \in M_{m \times n}(F)$ and $x \in F^n$,

$$\|Ax\|_p \leq \|A\|_p \|x\|_p$$

Prop. Given $A \in M_{m\times n}(F)$ where $F = \mathbb{R}$ or $\mathbb{C}$,

$$\|A\|_1 = \max_{1\le j\le n} \sum_{i=1}^{m} |A_{ij}| \quad \text{and} \quad \|A\|_\infty = \max_{1\le i\le m} \sum_{j=1}^{n} |A_{ij}|$$

Proof. Given $x \in \mathbb{R}^n$ with $\|x\|_1 = 1$, denote $x = (x_1, \cdots, x_n)$.

$$\|Ax\|_1 \underset{\text{def}}{=} \left\| \sum_{j=1}^{n} x_j \cdot Ae_j \right\|_1 \underset{\text{Homo + Sub.add}}{\le} \sum_{j=1}^{n} |x_j| \cdot \|Ae_j\|_1 \le \sum_{j=1}^{n} |x_j| \cdot \max_{1\le j\le n} \|A\cdot e_j\|_1 = \max_{1\le j\le n} \|Ae_j\|_1.$$

And, take $x = e_k$ such that e_k maximizes the $\|Ae_j\|$, then we obtain the first.

Given $x \in \mathbb{R}^n$ with $\|x\|_\infty = 1$. Then, for any $1 \le k \le m$,

$$|(Ax)_k| = \left| \sum_{j=1}^{n} A_{kj} \cdot x_j \right| \le \sum_{j=1}^{n} |x_j| \cdot |A_{kj}| \le \sum_{j=1}^{n} |A_{kj}|$$

and the equality holds when $x = (1, 1, \cdots, 1)$. So proved. ☒

The $\|\cdot\|_2$ is so special, because it is induced by the inner product.
Note that the fact that: for any unitary $Q \in M_{n\times n}(F)$, Q preserves the norm. That is,

$$\|Qx\|_2^2 = \langle Qx, Qx\rangle = \langle x, x\rangle = \|x\|_2^2$$

Thm. Given $A \in M_{m\times n}(F)$ where $F = \mathbb{R}$ or $\mathbb{C}$,

$$\|A\|_2 = \sigma_{max}, \text{ the largest singular value of } A.$$

Proof. Given A, there is a singular value decomposition,

$$A = U\Sigma V^* \text{ where } U \in M_{m\times m},\ V \in M_{n\times n} \text{ are unitary,}$$

and $\Sigma = \begin{pmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{pmatrix} \in M_{m\times n}$, $\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_r > 0$

So,

$$\|Ax\|_2 = \|U\Sigma V^* x\|_2 = \|\Sigma V^* x\|_2 = \|\Sigma y\|_2 \text{ where } y = V^* x$$

Since the unitary operator is isomorphism, and isometry, we can replace V^*x into y.
And, the p-norm of diagonal matrix is the largest absolute value of its entry,
so proved.

「Problems.

For $A = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 2 \end{bmatrix}$, Compute $\|A\|_1, \|A\|_\infty, \|A\|_F, \|A\|_2$.

Proof.

$\|A\|_1 = |-2| + |1| = 3$, $\|A\|_\infty = |1| + |-2| = 3$.

$AA^* = \begin{bmatrix} 5 & -2 \\ -2 & 5 \end{bmatrix}$

$\|A\|_F = \sqrt{1+4+1+4} = \sqrt{10}$. $\left(= \sqrt{tr(AA^*)}\right)$.

Meanwhile, $A^*A = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 2 \\ 0 & 2 & 4 \end{bmatrix}$ has a characteristic polynomial

$\chi_{A^*A} = t(t-3)(t-7)$

So, $\|A\|_2 = \sqrt{7}$.